

Student Management System - Characteristics

1. User Authentication System

- Secure login and logout functionality.
- Only authenticated users can access the system.
- Each user can manage only their own student data.

2. CRUD-Based Student Management

- Add new students (Create).
- View student list (Read).
- Edit student details (Update).
- Delete student records (Delete).

3. Relational Database Design

- Students linked to users using ForeignKey.
- Marks linked to students using ForeignKey.
- Ensures structured and organized data storage.

4. Marks Management System

- Add multiple subjects for each student.
- Edit and delete marks.
- Maintain subject-wise records for each student.

5. Performance Analytics

- Calculate total marks.
- Compute percentage automatically.
- Identify top-performing students.

6. Dashboard

- Displays total number of students.
- Shows total marks entries.

- Calculates average marks.
- Highlights top performer.

7. Search & Filtering

- Search students by name.
- Search students by course.
- Improves data retrieval efficiency.

8. Pagination System

- Displays limited records per page.
- Improves performance for large datasets.
- Enhances user experience.

9. User Feedback System

- Displays success messages for actions.
- Provides feedback for add, edit, and delete operations.
- Improves user interaction.

10. Modular Architecture

- Uses Django MVC pattern.
- Separates models, views, and templates.
- Ensures maintainability and scalability.

11. Responsive UI (Bootstrap)

- Modern and clean interface.
- Uses Bootstrap components.
- Mobile-friendly design.

12. Security & Data Isolation

- Users can access only their own data.
- Protected views using login_required.
- Prevents unauthorized access.

